

Euclid's Division Lemma

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1 Statement

Given $A > 0$ and a divisor $B > 0$, there exists a unique solution (C, D) with quotient $C > 0$ and remainder $D > 0$ to the equation $A/B = C + D/B$ given the condition $0 \leq D < B$, where we say C is the quotient and D is the remainder.

2 Proof 1

For example, consider the following:

$$\frac{A}{B} = C + \frac{D}{B} \text{ as}$$

$$\frac{1}{4} = 0 + \frac{1}{4}$$

$$\frac{2}{4} = 0 + \frac{2}{4}$$

$$\frac{3}{4} = 0 + \frac{3}{4}$$

$$\frac{4}{4} = 1 + \frac{0}{4}$$

$$\frac{5}{4} = 1 + \frac{1}{4}$$

$$\frac{6}{4} = 1 + \frac{2}{4}$$

$$\frac{7}{4} = 1 + \frac{3}{4}$$

$$\frac{8}{4} = 2 + \frac{0}{4}$$

Then you can see for each A, B, C there is only one D such that $0 < D \leq B$.

This is equivalent to $A = B \cdot C + D$ given A, B and $0 < D \leq B$ having a unique solution C, D .

Finally, this is equivalent to $A/B = C + D/B$ given A, B and $0 < D \leq B$ having a unique solution C, D .

3 Proof 2

Consider the sequence $0B < 1B < 2B < 3B < 4B < 5B < \dots$

Either $A = C \cdot B$ for some C (case 1)

or

$C \cdot B < A < (C + 1) \cdot B$ for some C . (case 2)

We see that the value of C is unique.

In the first case $A = C \cdot B + 0$ meaning $A/B = C + D/B$ where $D = 0$.

In the second case $A = C \cdot B + D$ and since A is somewhere between an integer multiple of B , then $0 \leq D < B$, meaning $A/B = C + D/B$ where $0 \leq D < B$.

4 Proof 3

Given $B > 0$, each pair of integers (C, D) such that $0 \leq D < B$ is associated with a unique integer $B \cdot C + D$

I demonstrate below by construction:

$$\mathbb{Z} = \{ \dots, \quad B \cdot (-1) + 0, \quad B \cdot (-1) + 1, \quad \dots, \quad B \cdot (-1) + (B-1), \\ B \cdot (0) + 0, \quad B \cdot (0) + 1, \quad \dots, \quad B \cdot (0) + (B-1), \\ B \cdot (1) + 0, \quad B \cdot (1) + 1, \quad \dots, \quad B \cdot (1) + (B-1), \quad \dots \}$$

Notice, that the map between (C, D) and $B \cdot C + D$ is a bijection and that $B \cdot C + D$ covers every integer in $\mathbb{Z}$.

So, given $B > 0$, for any $A \in \mathbb{Z}$ there exists a unique pair (C, D) , such that $0 \leq D < B$ and $A = B \cdot C + D$

5 Proof 4

From George Andrews.

$A/B = C + D/B$ has unique C and unique D where $0 \leq D < B$ if

$A = BC + D$ has unique C and unique D where $0 \leq D < B$.

We can use the basis representation theorem to see that

$$A = C_1 \cdot B^N + C_2 \cdot B^{N-1} + \dots + C_{N-1} \cdot B + C_N$$

with each $0 \leq C_i < B$.

If we write C_n as “ D ” and $C_1 \cdot B^{N-1} + C_2 \cdot B^{N-2} + \dots + C_{N-2}$ as “ C ”, then we have

$A = BC + D$, where C is unique since the basis representation for base B gives $C_1, C_2, \dots, C_{N-1}$ unique and therefore $C = C_1 \cdot B^{N-1} + C_2 \cdot B^{N-2} + \dots + C_{N-2}$ unique and basis representation for base B gives $D = C_n$ unique and $0 \leq D < B$.

6 Note

Given A, B ,

$A = B \cdot C + D$ has one solution with $0 \leq D < B$.

Given A, C ,

$A = B \cdot C + D$ has one solution with $0 \leq D < C$.

Given A, B, C ,
 $A = B \cdot C + D$ has one solution with $0 \leq D < \max(B, C)$.
In other words, given an addition constraint, in order to guarantee a solution,
we need to relax our other constraint on the inequality D .